

Research Assessment Exercise 2026

Impact Case Study

University: The University of Hong Kong (HKU)
Unit of Assessment (UoA): 13 – Computer studies/science (incl. information technology)
Title of case study: UTEE (Ubiquitous Trusted Execution Environments): A Secure and High-performance Big-data Computing Infrastructure System on Public Clouds
Is this case study continued from a case study submitted in RAE 2020? N
(1) Summary of the impact (indicative maximum 100 words) Since 2020, Prof Cui has addressed three limitations in Trusted Execution Environments (TEE). Ubiquitous TEE (UTEE) developed four software protocols which enhance the performance and efficiency of TEE systems, processing datasets that are three times larger. UTEE is the essential system in Huawei's secure cloud computing service TICS. In 2021, Huawei estimated that TICS reached 3 million active cloud tenants in 170 countries. In 2020, TICS was endorsed by the Chinese government. Huawei integrated UTEE into Huawei Cards that was released by China Citic Bank in 2021, with over 2 million users reported in September 2023. UTEE protects users' information in credit card transactions, eliminating fraud and data breach risks. Huawei Cards users exceeded 3.2 million in September 2024.
(2) Underpinning research (indicative maximum 500 words) With the growth of big data and cloud-based applications, users can face data security vulnerabilities and risks of data breaches. The "Trusted software computing base" (TSCB) is defined as the common collection of "trusted software" that must be assumed to be completely secure by any cloud vendors and users. This includes the Linux operating system which comprises over 40 million lines of code and has been written by over 1500 organisations globally. Bugs in the operating system can easily leak user data from the cloud to the internet, known as a data breach. To avoid such breaches, a hardware-based security entity called the Trusted Execution Environment (TEE) has become available for commercial central processing units (CPUs). The TEE is physically isolated from the rest of the CPU and is able to ensure the confidentiality and integrity of big-data, reducing the risk of data breaches. The hardware underlying the TEE offers protection from the software environment and people with high levels of privilege in the system, such as administrators. All existing TEE systems have three significant limitations which remain despite significant efforts to date by academia and industry: - TEE remains a low-level programming environment with a large TSCB, which means that it cannot be used easily by regular cloud software developers (who often use Java); - TEE systems cannot support typical large-scale datasets (e.g. terabytes) as they suffer from significant performance overheads (20-100-fold) compared to native and insecure big data computation; - TEE systems are only available for CPUs and not for more powerful graphics processing units (GPUs). Prof Heming Cui and his team at HKU have undertaken research since 2020 which directly addresses these three weaknesses in existing TEE systems. The Ubiquitous TEE (UTEE) project has successfully developed four novel secure software protocols which together have greatly enhanced the performance and efficiency of TEE systems. Cronus was developed in 2021 and was the first time that a CPU's TEE was extended so that it could also work on more powerful GPUs [4.1]. UTEE has a low overhead in execution time (7.1%), compared to the native and insecure systems previously used for GPU computations. This therefore addresses one of the previous weaknesses of TEE systems in operation to date. The research focused on developing four novel secure software protocols. By using a micro-kernel operating protocol, the TEE security guarantees were extended and this enabled the TEE system to be extended to GPUs for the first time [4.1, 4.3].

Uranus [4.2] concentrated on the Trusted Software Computing Base (TSCB), with a TEE-based dependent code verification and attestation protocol. This enables UTEE to run Java (a widely used high-level programming language in clouds) and to use a garbage collection protocol that greatly reduces memory usage and supports large-scale datasets on CPUs. It achieved a 79% reduction in the size of the TSCB compared to UTEE's competitors, from 1050K to 250K lines of code, which reduced the performance overhead. This reduction results in UTEE being able to handle and process datasets that are three orders of magnitude larger than those supported by competitors such as UC Berkeley, Microsoft, AWS and Google. The technologies and protocols developed in 4.1, 4.2 and 4.3 are combined and integrated in 4.4, demonstrating that the UTEE project has achieved three significant goals, namely that it has: - Greatly reduced the size of the TSCB of the world's best cloud vendors; - Significantly improved the performance of TEE systems; and - Supported the introduction of TEE in general mainstream GPUs. The work was supported by RGC GRF grant funding, with work from 2019-22 focused on the development of new systems and algorithms to preserve the privacy of big data in clouds [G2]. Since 2024, work has been ongoing on the use of micro-kernel systems and algorithms to enable secure, reliable high performance on public clouds [G1]. Prof Heming Cui and his team have worked in partnership with Huawei to commercialise the UTEE technology which is used in their secure cloud computing service. UTEE is integrated into commercial products offered by Huawei which deliver improved information security for millions of customers.
(3) Period when the underpinning research was undertaken: The underpinning research was undertaken by Prof Cui at HKU between 2020 and 2025. Research is ongoing through the funded research grant.
(4) References to the research (indicative maximum of six references) HKU PhD students, who are primarily supervised by Prof. Heming Cui, are marked with '___'. All published papers [4.1, 4.2, 4.3, 4.4] are in highly regarded publications, as ranked by Google Scholar (on security or computing systems). All papers were published in top peer reviewed venues. Research grants were awarded following competitive peer review. [4.1 Cronus ACM Micro 2022] <u>Jianyu Jiang</u>, <u>Qi Ji</u>, <u>Tianxiang Shen</u>, <u>Xusheng Chen</u>, <u>Shixiong Zhao</u>, Sen Wang, Li Chen, Gong Zhang, Xiapu Luo, and Heming Cui, CRONUS: Fault-isolated, Secure and High-performance Heterogeneous Computing for Trusted Execution Environments. [4.2 Uranus ACM AsiaCCS 2020] <u>Jianyu Jiang</u>, <u>Xusheng Chen</u>, Tzs On Li, <u>Cheng Wang</u>, <u>Tianxiang Shen</u>, <u>Shixiong Zhao</u>, Heming Cui, Co-Li Wang, and Fengwei Zhang, Uranus: Simple, Efficient SGX Programming and Its Applications. [4.3 Sotter Usenix ATC 2022] <u>Tianxiang Shen</u>, <u>Ji Qi</u>, <u>Jianyu Jiang</u>, Xian Wang, Siyuan Wen, <u>Xusheng Chen</u>, <u>Shixiong Zhao</u>, Sen Wang, Li Chen, Xiapu Luo, Fengwei Zhang, and Heming Cui, SOTER: Guarding Black-box Inference for General Neural Networks at the Edge. [4.4 ACM TDSC 2021 (journal)] <u>Tianxiang Shen</u>, <u>Jianyu Jiang</u>, Yunpeng Jiang, <u>Xusheng Chen</u>, <u>Ji Qi</u>, <u>Shixiong Zhao</u>, Fengwei Zhang, Xiapu Luo, and Heming Cui, DAENet: Making Strong Anonymity Scale in a Fully Decentralized Network. [G1 RGC GRF grant] PI (Heming Cui), "Micro-kernel Inspired Systems and Algorithms (MISA): Enabling Secure, Reliable and High-performance Micro-services on Public Clouds", Hong Kong RGCGRF (Ref HKU17208223), HK \$1.1 million, 2024-27 [G2 RGC GRF grant] PI (Heming Cui), "New Systems and Algorithms for Preserving Big-data Privacy in Clouds", Hong Kong RGCGRF (Ref HKU17202318), HK \$1.7 million, 2019-22
(5) Details of the impact (indicative maximum 750 words) In 2022, China represented 13% of global cloud infrastructure spend, according to Canalys, with cloud infrastructure services expenditure in mainland China growing 21% year on year to reach US \$7.3 billion in Q1 2022, an increase of US \$1.3 billion from the same period in 2021 [S1]. Huawei Cloud was the second-largest cloud service provider in Q1 2022, growing 11% to take an 18.0% market share [S1]. The 2022 China Big Data Platform Market Shares IDC report

states that Huawei Cloud was ranked first among public cloud services for big-data platform market share in China. Its 2022 China market share is greater than its three closest competitors combined, with an annual growth rate of 53.1%, the highest of all vendors in China [S2].

The research undertaken by Prof Cui and his team has directly contributed to multiple impacts:

Commercialisation of the UTEE technology and licensing income

Prof Cui and his team have developed a close collaboration with Huawei through joint publications on secure systems and the outputs about the UTEE technology are referenced as underpinning research to this case. The academic prototype of UTEE has been further developed by Prof Cui and his team with Huawei to make it suitable for large-scale industry deployment by Huawei in China and the world. The open-source academic prototype of UTEE is available to academic researchers and industry globally.

During the RAE period, HKU and Huawei have jointly applied for 15 patents, including two in the USA and 13 in China, seven of which are strongly related to UTEE [S3, S5, p.1], and these protect the IP rights of both parties. The HKU Technology Transfer Office transferred the patents to Huawei between 2020 and 2025, resulting in licensing fees paid to the HKU of HK \$7.14 million [S4]. This is very rare for security or systems research in Hong Kong.

UTEE technology used in Huawei secure cloud computing service

Prof Heming Cui has negotiated an exclusive license with Huawei, with the UTEE technology being made available to its cloud and credit card users and to its business customers (such as banks). In 2021, Huawei estimated that UTEE technology reached 3 million active cloud tenants (including business customers and individual users) located in 170 countries [S5, p.1].

The industry-grade version of UTEE has been jointly developed by Prof Cui and Huawei to ensure that the software reaches the quality of industry-grade deployment for global users. In April 2021, UTEE was commercialised and “successfully deployed in TICS as a core system component” [S5, p.1]; in TICS, “UTEE is invoked to protect the confidentiality of every big-data computing task on plaintext user data” [S5, p.2].

The big data product evaluation of the China government (Academy of Information and Communications Technology) is an authoritative big data evaluation system in China, and the evaluation scope covers the basic capabilities and performance of big data products. In December 2020, in an evaluation test TICS “passed 100% of the 21 mandatory use cases at one time, and also passed the four optional use cases at one time, handing over a perfect answer” [S6]; significantly, only about two relevant software systems have been endorsed at this quality level by China since 2020. This endorsement as a trusted and reliable secure software meets one of the world’s most stringent user data security laws (The Data Security Law of China). The TSCB in UTEE is 79% smaller than for worldwide competitor systems, as outlined in [4.2]. China government’s endorsement makes UTEE much easier for UTEE to be checked and endorsed by diverse countries’ governments worldwide.

UTEE integrated into commercial products offered by Huawei, delivering improved information security for customers

According to the notorious, publicly released “IBM’s Cost of a Data Breach Report in 2024”, the financial loss incurred globally by every data breach incident is USD \$4.88 million on average. Ironically, in April 2021, Minnesota University’s Linux Hypocrite Commits caused the injections and fixes of 190+ known bugs or incidents in Linux. It is possible that many unknown bugs injected by this organisation will still exist in the Linux services offered by many global public cloud vendors. This means that there is a greater risk of data breaches or user data being compromised or mis-used.

Huawei has worked with global strategic business partners and has integrated UTEE (TICS) into the Huawei Cards that was released by China Citic Bank in 2021 [S8]. The system protects the confidentiality of users’ plaintext information by placing it in a trusted zone so that it cannot be seen by anyone else. The protection applies to the data in every credit card transaction and all but eliminates the risk of fraud or a data breach against the OS. Citic Bank recorded more than 2 million users of Huawei Cards in September 2023 (S8). Citic Bank publicly announced an

annual estimated total transaction amount of CNY \$26289 per user and a total annual financial profit of CNY \$667 per user (S7). Therefore, multiplying the per user figure by the 2 million Huawei Cards users equates to an annual profit of CNY \$1,334,000,000 for China Citic Bank alone. It has also been reported that Huawei Cards has been extended to other major banks, including Guangdong Development Bank who reported 1.2 million users in 2024 [S9]. Total numbers of Huawei Cards users therefore exceeded 3.2 million in September 2024 [S8, S9]. In addition, China UnionPay (equivalent to Mastercard/Visa) stated that on April 8 2020, Huawei launched a new service at the HUAWEI P40 series flagship mobile phone launch conference, namely the UnionPay standard Huawei Card [S10]. Due to protection of phone users' personal privacy data, it is not possible to quantify how many users have activated this service on their phones.

The Director of the Theory Lab at Huawei states that "Before the UTEE project, users' plaintext big-data computing tasks on public clouds face high risks on data breaches (e.g., by the 1500 organizations on jointly developing the Linux OS). Since the UTEE project, these risks are effectively tackled for the users." [S5, p.2].

Overall, the UTEE technology that Prof Cui and his team have developed with Huawei has led to significant improvements in data security and the performance of public cloud services. The beneficiaries include: the cloud vendors who deploy UTEE, notably Huawei; customers of the cloud vendors; and government(s) that endorse the deployment of UTEE in their countries.

(6) Sources to corroborate the impact (indicative maximum of 10 references)

S1 Cloud services spend in China hits US\$7.3 billion in Q1 2022, June 2022

<https://canalys.dev/newsroom/china-cloud-market-Q1-2022>

S2 China Big Data Platform Market Shares, 2022 report, IDC

<https://www.huaweicloud.com/intl/en-us/news/20230726145034918.html>

S3 15 patent applications of the UTEE project filed in the RAE review period, 2 in the USA and 13 in China <https://hemingcui.github.io/rae2026-utee/details-of-patents.pdf>

S4 Evidence of licensing fee paid by Huawei to Hong Kong University for the patents (with HKU official chop) <https://hemingcui.github.io/rae2026-utee/hku-licensing-fee.pdf>

S5 Testimonial endorsement letters from the Director of the Theory Lab of Huawei, with Huawei's official chops (First one endorsed in 2021; Second one endorsed in 2025), <https://hemingcui.github.io/doc/utee-ack.pdf>

S6 Endorsement of TICS by the China Academy of Information and Communications Technology, December 2020 <https://icode.best/i/87602637140406> (Chinese);

<https://hemingcui.github.io/rae2026-utee/china-endorsement.pdf> (English translation).

S7 The China Citic Bank's per user credit card statistics since 2020: (Chinese)

<https://m.eeo.com.cn/2021/0326/482900.shtml>; (English translation)

<https://hemingcui.github.io/rae2026-utee/china-citic-bank-credit-card-stats.pdf>.

S8 The Huawei Cards, implemented based on TICS for the security of all credit card transactions, announced by China Citic Bank in September 2021: (Chinese)

<https://huaweicloud.com/intl/zh-cn/news/2020/20210604100805740.html>; (English translation),

<https://hemingcui.github.io/rae2026-utee/huaweicards-china-citic-bank-annoucement.pdf>;

confirmation of this bank's Huawei Cards users of exceeding 2 million in September 2023; (Chinese) <https://www.cls.cn/detail/1473982>; (English translation)

<https://hemingcui.github.io/rae2026-utee/huawei-Citic-bank-card-2million-users.pdf>.

S9 Confirmation of 1.2 million users of Huawei Cards users for Guangdong Development Bank in 2024; (Chinese) <https://m.mp.oeeee.com/a/BAAFRD0000202412111033651.html>; (English translation)

<https://hemingcui.github.io/rae2026-utee/huaweicards-guangfa-bank-users.pdf>.

S10 China UnionPay (China's national credit card organization) released UnionPay Huawei Cards service on Huawei P40 mobile phones: (Chinese)

<http://baijiahao.baidu.com/s?id=1663432459663052187&wfr=spider&for=pc>; (English

translation) <https://hemingcui.github.io/rae2026-utee/huaweicards-china-unionpay-release.pdf>.